

PACKAGE INSERT

Brand or product name	Biamuline Oral Solution 12.5%
Name and Strength of Active Substance/ Preservative(s)	Composition per 1 ml: Tiamulin (as hydrogen fumarate).....125.00 mg (equivalent to 101.20 mg tiamulin) Antimicrobial Preservative (s): Benzyl alcohol20.00 mg Sodium methyl parahydroxybenzoate1.00 mg Sodium propyl parahydroxybenzoate.....0.20 mg
Product Description	Oral Solution. Clear colourless to pale yellow solution.
Pharmacodynamics	Tiamulin is a semisynthetic bacteriostatic antibiotic that belongs to the pleuromutilins class, acting at the ribosomal level by inhibition of bacterial proteins synthesis. Tiamulin showed in vitro activity against a large variety of bacteria including <i>Brachyspira hyodysenteriae</i>, <i>Brachyspira pilosicoli</i>, <i>Lawsonia intracellularis</i> and <i>Mycoplasma spp.</i> Tiamulin has a bacteriostatic activity at therapeutic doses and acts on the ribosomal fraction 70S, the primary binding site is on the 50S subunit with a potential secondary binding site between subunits 50S and 30S. It appears to inhibit the production of microbial protein by forming an initial biochemically inactive complex, which prevents the elongation of the polypeptide chain. The mechanisms responsible for the development of resistance to the class of pleuromutilins in <i>Brachyspira spp</i> would be based on ribosomal target site mutations. The development of a clinically visible resistance to tiamulin requires a combination of mutations around its binding site. Resistance to tiamulin may be associated with a decreased sensitivity towards other pleuromutilins.
Pharmacokinetics	Pig: Tiamulin is rapidly absorbed in pigs (over 90%) after oral administration and widely distributed in the body. Following oral administration of a single dose of 10 mg and 25 mg of tiamulin / kg body weight, the C_{max} reached 1.03 µg / ml and 1.82 µg / ml respectively by microbiological assay and the T_{max} is reached in 2 hours for both doses. Tiamulin is concentrated in the lungs, target tissue, as well as in the liver where it is metabolized and excreted (70-85%) by the bile, the residues are excreted by the kidneys (15-30%). Tiamulin which has not been absorbed goes through the intestine to the colon where it is

	concentrated. Hens: Tiamulin is well absorbed in chickens (70-95%) after oral administration. Tiamulin is widely distributed in the body and is concentrated in the liver and kidneys (excretion sites) and in the lungs (30 times the serum concentration). Excretion is principally done via the bile (55-65%) and kidney (15-30%), mainly as microbiologically inactive metabolites. It is relatively fast, 99% of the dose in 48 hours. Turkeys: In turkeys, serum concentrations of tiamulin are similar to those observed in chicken. In breeding, the mean serum concentration after administration of 0.025% tiamulin was 0.36 µg / ml (0.22 to 0.5 µg / ml).
Environmental Properties (for product used directly in the environment e.g. medicine for fish)	N/A
Indication	Diseases caused by bacteria susceptible to tiamulin. Pigs: - Treatment of hemorrhagic enteritis caused by <i>Brachyspira hyodysenteriae</i>. - Reduction of lesions and treatment of enzootic pneumonia caused by <i>Mycoplasma hyopneumoniae</i>, eventually associated with <i>Mycoplasma hyorhinis</i>, <i>Pasteurella sp.</i> or <i>Haemophilus sp.</i> Chicken: - Treatment and prevention of chronic respiratory disease (CRD) and airsacculitis caused by <i>Mycoplasma gallisepticum</i> and <i>Mycoplasma synoviae</i>. In turkeys: - Treatment and prevention of infectious sinusitis and airsacculitis caused by <i>Mycoplasma gallisepticum</i>, <i>Mycoplasma meleagridis</i> and <i>Mycoplasma synoviae</i>.
Recommended Dosage	Pigs: 4.05 to 8.10 mg of tiamulin per kg of body weight per day, administered in the drinking water for 5 consecutive days, depending on the severity of the infection and / or disease duration or 4-8 ml per 100 kg of bodyweight per day for 5 days. During the preparation of large quantities of medicated water, prepare a first concentrated solution and dilute until the desired

	concentration. To ensure the correct dose, the body weight should be accurately determined to avoid under-dosing. Prescribed doses must be measured with suitable equipment. Renew medicated water every 24 hours. Chicken (broilers, replacement pullets, laying breeding hens): 20.2 mg of tiamulin per kg of body weight per day, administered for 3 to 5 consecutive days. The dosage is normally achieved by an incorporation rate of 20 ml of solution per 100 kg in the drinking water. Turkeys (poults, breeding turkeys): 32.4 mg of tiamulin per kg of body weight per day, administered for 3 to 5 consecutive days. The dosage is normally achieved by an incorporation rate of 32.4 ml of solution per 100 kg in the drinking water. The quantity of medicated drinking water consumed by animals depends on their physiological and clinical state. To obtain the recommended dosage, the concentration of tiamulin should be therefore adjusted. Preventive treatment with tiamulin should only be initiated after confirmation of infection by <i>M. gallisepticum</i>, <i>M. synoviae</i> and <i>M. meleagridis</i> and as an help in the context of a prevention strategy to reduce the clinical signs and mortality from respiratory disease in breedings where infection of the ovum is likely because the disease is known to exist in the previous generation. The prevention strategy should include efforts to eliminate the infection from the previous generation.
Mode of Administration	Orally administration diluted in drinking water or milk.
Contra-indications	Animals should not receive products containing ionophores such as monensin, narasin or salinomycin during treatment with tiamulin or for at least seven days before or after treatment with tiamulin. A significant slowdown in growth or death may occur. See section 4.8. for information about the interaction between tiamulin and ionophores.
Warnings and Precautions	a) Special warnings for each target species In case of reduction of water consumption, the incorporation rate in water may be increased to achieve the target dose. Animals which do not drink enough must be treated by parenteral route. Do not use in case of hypersensitivity to the active substance or to one of the excipients. Do not use in case of resistance to tiamulin. b) Special precautions for use

	i) Special precautions for use in animals The use of the product should be based on susceptibility testing and take into account official and local policies for the use of antibiotics. Improper use of the product may increase the prevalence of bacteria resistant to tiamulin and may decrease the effectiveness of treatment using substances related to tiamulin. The repeated use or long term use of this drug should be avoided by improving breeding conditions, cleaning and disinfection. In the absence of satisfactory response to treatment, the diagnosis should be reconsidered. See section 4.8 for information about the interaction between tiamulin and ionophores. ii) Special precautions to be taken by the person administering the medicinal product to animals When mixing the veterinary medicinal product, direct contact with eyes, skin and mucous membranes should be avoided. Personal protective equipment should be worn when mixing the veterinary medicinal product or handling of the medicated water: a coverall, waterproof gloves, a disposable half-face respirator in accordance with European standard EN 149 or a non-disposable respirator in accordance with European standard EN 140 with European standard EN 143 filter. Wash exposed skin. In case of accidental ingestion, seek medical advice immediately and show the instruction for use or the product label. People with known hypersensitivity to tiamulin should handle the product carefully. iii) Other precautions To reduce the risk on aquatic environment, maintain a spreading distance of 10 meters from waterways.
Interactions with Other Medicaments	For hens and turkeys: Tiamulin interacts with ionophores such as monensin, salinomycin and narasin which can cause indistinguishable signs of ionophores intoxication. Animals should not receive products containing ionophores during treatment with tiamulin or for at least seven days before or after treatment with tiamulin. A significant slowdown of the growth, ataxia, paralysis or death may occur. For this purpose, it must be notified to the supplier of food that tiamulin will be used and that these products should not be incorporated into or contaminate the food. Check for the

	presence of any ionophore in the food in case of suspicion. In the event of interaction, immediately stop the treatment of drinking water containing tiamulin and replace the water with clean water. Remove contaminated food as soon as possible and replace with a food that does not contain incompatible product with tiamulin.
Statement on usage during pregnancy and lactation	The drug can be used in pigs during pregnancy or lactation. The drug can be used in laying and breeding hens and turkeys.
Adverse Effects/Undesirable Effects	In rare cases, erythema or slight oedema of the skin may occur in pigs after treatment.
Overdose and Treatment	Pigs: single oral doses of 100 mg / kg body weight in pigs cause hyperventilation and abdominal discomfort. With a dose of 150 mg / kg, no effect on the central nervous system were observed, apart from sedation. With a dose of 55 mg / kg administered for 14 days, a transitional salivation and slight gastric irritation were observed. The minimum lethal dose has not been established in pigs. Poultry: The LD50 is 1290 mg / kg body weight in chicken and 840 mg / kg body weight in turkey. Clinical signs of acute toxicity in chicken are vocalization, clonic cramps and lateral decubitus. In turkey, signs of acute toxicity include clonic cramps, the lateral decubitus or supine, salivation and ptosis. In case of intoxication signs, immediately remove the medicated water, replace with the non-medicated drinking water, and apply a supportive symptomatic treatment.
MRL	Tiamulin (as hydrogen fumarate):- Pig: Muscle - 100 µg/kg Liver - 500 µg/kg Hen: Muscle, skin + fat - 100 µg/kg Liver - 1000 µg/kg Eggs - 1000 µg/kg Turkey: Muscle, skin + fat - 100 µg/kg Liver - 300 µg/kg
Incompatibilities(for injections only)	N/A
Withdrawal period(s)	Pigs and turkeys: - Meat and offal: 4 days. Chickens:

	- Meat and offal: 1 day. - Eggs: Zero days.
Storage conditions (may be omitted if the information is stated on the label or outer carton labels)/(should be supported by stability data)	Store at 25°C/60% Relative Humidity.
Dosage forms and packaging available	Dosage form: Oral Solution Packaging available: Flask of 1 litre Canisters of 5 litre
Name and address of manufacturer	Laboratoires biové 3, rue de Lorraine, 62510 Arques, France
Name and Address registration holder	Weissen Company (M) Sdn Bhd Lot 105, No. 84, Jalan Velox 23, Taman Velox, 48000 Rawang, Selangor, Malaysia
Date of revision of Package insert	01.07.2024
Shelf-life of the veterinary product as packaged for sale	36 months
Shelf-life after first opening the container (where relevant)	3 months
Shelf-life after first dilution or reconstitution according the directions (where relevant)	24 hours